On my honor I have neither given nor received any unauthorized aid or inappropriate assistance for all sessions of this exam. The work done on this exam is entirely my own. I understand that by the actual code, violations of these principles will lead to zero grade and is subject to both discipline issues.

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Q1) $(1 \quad 0 \quad -2 \quad 1) * (0 \quad 1 \quad -1 \quad 0 \; 1)$

$1\ 0\ \underline{1}\ \underline{0}\ -2\ 1\ ]\ -1 \quad n=0$
$\quad\ -1\ \underline{1}\ 0$

$1\ 0\ -1\ \underline{1}\ \underline{0}\ -2\ 1\ ]\ 1 \quad n=-1$
$\quad\quad\ 1\ \underline{0}$

$1\ \underline{0}\ -2\ 1\ ]\ -2 \quad n=1$
$1\ 0\ \underline{-1}\ 1\ 0$

$1\ \underline{0}\ -2\ 1\ ]\ 4 \quad n=2$
$1\ \underline{0}\ -1\ 1\ 0$

$1\ \underline{0}\ -2\ 1\ ]\ -1 \quad n=3$
$\quad\ \underline{1}\ 0\ -1\ 1\ 0$

$1\ \underline{0}\ -2\ 1\ ]\ -2 \quad n=4$
$\quad\ \underline{0}\ 1\ 0$

$1\ \underline{0}\ -2\ 1\ ]\ 1 \quad n=5$
$\quad\ \underline{0}\ 0\ 1$

result:

1	$\underline{-1}$	-2	4	-1	-2	1

Q2) $\frac{dy(t)}{dt} + 4y(t) = 7e^{-3t}u(t)$ $\quad Y(w) = ?$

FT $\qquad$ FT

$FT[e^{-\alpha t}u(t)] = \frac{1}{\alpha + jw}$

$jwY(w) + 4Y(w) = FT[7e^{-3t}u(t)]$

$$7\int_{-\infty}^{\infty} e^{-3t}u(t)\cdot e^{-jwt}dt = 7\cdot\int_{0}^{\infty} e^{t(-3-jw)}dt$$

$$= 7\cdot \left.\frac{e^{(-3-jw)t}}{-3-jw}\right|_0^{\infty}$$

right side

$$= \frac{e^{-(3-jw)\infty}}{-3-jw} - \left(\frac{e^{(-3-jw)0}}{-3-jw}\right) = \frac{7}{3+jw}$$

$$\underbrace{jwY(w)}_{(3+jw)} + \underbrace{4Y(w)}_{(3+jw)} - \frac{7}{3+jw} = 0$$

$$(3jw - w^2)Y(w) + 12Y(w) + 4jwY(w) - 7 = 0$$

$$-3jwY(w) + w^2Y(w) + 12Y(w) + 4jwY(w) + 7 = 0$$

$$w^2Y(w) - 7jwY(w) - 12Y(w) + 7 = 0$$

$$(w^2 - 7jw - 12)Y(w) + 7 = 0$$

$$Y(w) = \frac{-7}{w^2 - 7jw - 12}$$

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